IK2215: Network Design Report

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1 General Information

ASN: 114 **NETWORK:** 1.114.0.0/20

2 Network overview

This section contains an overview of network design.

2.1 Network diagram

The network design is illustrated in Figure 1 below.

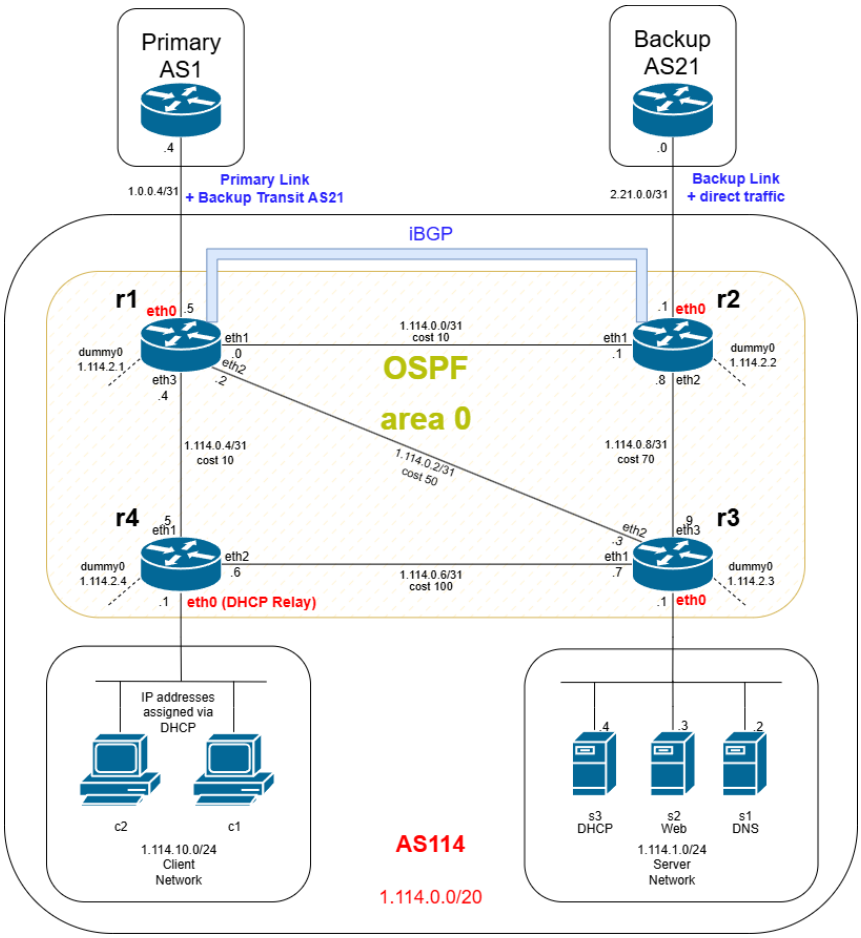


Figure 1: Network Diagram

2.2 IP address allocation

The IP addresses and domain names of all device interfaces are shown in the table below.

Device	Interface	IP address	Domain name
r1	eth0	1.0.0.5/31	eth0r1.isp114.lab
r1	eth1	1.114.0.0/31	eth1r1.isp114.lab
r1	eth2	1.114.0.2/31	eth2r1.isp114.lab
r1	eth3	1.114.0.4/31	eth3r1.isp114.lab
r1	dummy0	1.114.2.1/32	dummy0r1.isp114.lab
r2	eth0	2.21.0.1/31	eth0r2.isp114.lab
r2	eth1	1.114.0.1/31	eth1r2.isp114.lab
r2	eth2	1.114.0.8/31	eth2r2.isp114.lab
r2	dummy0	1.114.2.2/32	dummy0r2.isp114.lab
r3	eth0	1.114.1.1/24	eth0r3.isp114.lab
r3	eth1	1.114.0.7/31	eth1r3.isp114.lab
r3	eth2	1.114.0.3/31	eth2r3.isp114.lab
r3	eth3	1.114.0.9/31	eth3r3.isp114.lab
r3	dummy0	1.114.2.3/32	dummy0r3.isp114.lab
r4	eth0	1.114.10.1/24	eth0r4.isp114.lab
r4	eth1	1.114.0.5/31	eth1r4.isp114.lab
r4	eth2	1.114.0.6/31	eth2r4.isp114.lab
r4	dummy0	1.114.2.4/32	dummy0r4.isp114.lab
s1	eth0	1.114.1.2/24	ns.isp114.lab
s2	eth0	1.114.1.3/24	www.isp114.lab
s3	eth0	1.114.1.4/24	dhcpd.isp114.lab
c1	eth0	1.114.10.10/24	c1.isp114.lab
c2	eth0	1.114.10.11/24	c2.isp114.lab

3 Routing and service implementation

This section describes ISP implementation to realize routing and service requirements.

3.1 Routing

This section describe ISP implementation to fulfill routing requirements.

3.1.1 Intra-domain routing

We chose Open Shortest Path First (OSPF) because it is an interior gateway protocol that is well suited for ISP network environments. The protocol provides dynamic routing and fast convergence, and routing paths can be influenced by adjusting OSPF costs.

OSPF is enabled on all interfaces that connect the routers within the ISP, as well as on the logical dummy0 interfaces on each router. OSPF is also enabled on the server and client networks, but these interfaces are configured as passive. This prevents the formation of OSPF adjacencies with unauthorized or rogue routers in those networks.

To ensure deterministic routing paths and to avoid ECMP within the ISP, OSPF costs are modified on three links:

- r1 <-> r3 is assigned a cost of 50
- r2 <-> r3 is assigned a cost of 70
- r3 <-> r4 is assigned a cost of 100

All other links have a cost of 10, which is the default value. This configuration ensures that the ISP uses deterministic paths without ECMP, while still maintaining backup paths in case of a link failure.

Since only r1 and r2 run iBGP within the ISP, external network reachability must be announced to r3 and r4 through the IGP (OSPF). Redistributing all BGP routes into OSPF would achieve this, but that practice is strongly discouraged. Instead, default routes are injected into OSPF from both r1 and r2.

To ensure traffic prefers the primary path, metrics are adjusted so that the default route from r1 is preferred over the one from r2. As a result, r1 serves as the primary exit point, while r2 functions as a backup to avoid suboptimal routing.

To avoid a routing loop when the connection between r1 and r2 is brought down, we also redistribute AS21 prefixes into the OSPF routing domain on r2. Without this, r3 and r4 would send all traffic to r1, and then r1 would send it back because the AS21 prefix is supposed to use r2 as the path. This ensures that r3 and r4 always send the AS21 prefix directly to r2.

Table 1 and Table 2 below show the primary and secondary routing paths respectively.

Path	**r1**	**r2**	**servers**	**clients**
r1	X	-	r3	r4
r2	-	X	r1 r3	r1 r4
servers	r3	r3 r1	X	r3 r1 r4
clients	r4	r4 r1	r4 r1 r3	X

Table 1: Intermediate nodes in the primary routing path from row to column. X represents a path to itself, - represents a direct link without any intermediate node.

Path	**r1**	**r2**	**servers**	**clients**
r1	X	r3	r2 r3	r3 r4
r2	r3	X	r3	r3 r4
servers	r3 r2	r3 r1	X	r3 r4
clients	r4 r3	r4 r3	r4 r3	X

Table 2: Intermediate nodes in the secondary routing path from row to column (when the primary routing path fails). X represents a path to itself, - represents a direct link without any intermediate node.

3.1.2 Inter-domain routing

In our setup, r1 establishes an eBGP session with AS1 (the primary provider), while r2 establishes an eBGP session with AS21 (the backup provider).

To ensure consistency across AS114, we configure an iBGP session between r1 and r2 using their dummy0 interfaces. This ensures that the iBGP session remains up even if physical link between r1 and r2 fails. On this iBGP peering, the option next-hop self is applied so that internal routers do not need to learn about the external DMZ links. As a result, internal routers such as r3 and r4 rely purely on OSPF to reach either r1 or r2, while the actual exit point is selected according to BGP policy.

For outbound traffic, r1 is configured as the primary exit and r2 as the secondary. On r1, routes learned from AS1 are assigned a local preference of 200, making AS1 the preferred provider under normal conditions. On r2, routes to AS21's own prefix (2.21.0.0/20) are assigned a higher local preference of 300, ensuring that traffic destined for AS21 always exits directly via the r2–AS21 link. Other routes learned from AS21 retain the default local preference of 100, meaning they are only used if AS1 becomes unavailable.

For inbound traffic, only the aggregate prefix 1.114.0.0/20 is advertised to both AS1 and AS21. On r1, this aggregate is announced without AS-path prepending, except for the AS21 prefix (2.21.0.0/20),

which is prepended three times with our ASN to discourage AS21-bound traffic from using r1. This ensures that odther ASes prefers the direct to AS21 while still providing a backup path through AS114. On r2, all prefixes are prepended once. As a result, AS21 sees a shorter AS-path (length 2) via r2 and a longer AS-path (length 3) via r1, ensuring it prefers the direct backup link. Other external ASes continue to prefer the primary path via r1, as it offers the shortest AS-path to them.

3.2 Internet service

This section describes ISP implementation to fulfill service requirements.

3.2.1 DNS

The DNS server is assigned the IP address 1.114.1.2/24 and the hostname ns.isp114.lab. BIND is deployed as the DNS service, with a zone defined for isp114.lab. Within this zone, A records are created for all devices listed in the IP address allocation table.

A reverse lookup zone is also configured to enable IP-to-name resolution, consistent with the same allocation table. This reverse zone includes PTR records for all IP addresses within the ISP.

All clients, servers, and routers in the AS are configured to use 1.114.1.2 as their DNS resolver. This ensures that they query ns.isp114.lab for name resolution. To enable external ASes to resolve names under isp114.lab, a delegation is registered in the parent .lab zone, which is managed by the DNS server at 2.0.1.2 in AS2.

Finally, BIND is configured to forward queries it cannot resolve, such as DNS lookups for devices outside the ISP, to an upstream resolver, namely the root DNS server.

3.2.2 Web

The web server is deployed in the server network with the IP address 1.114.1.3/24 and the DNS name www.isp114.lab. It runs the Apache HTTP Server and hosts a simple website, which includes an index.html file containing the ASN, network details, and student information.

3.2.3 DHCP

The DHCP server is deployed in the server network with the IP address 1.114.1.4/24 and the DNS name dhcpd.isp114.lab. On this server, a DHCP scope of 1.114.10.10–1.114.10.200 is configured, and it runs the isc-dhcp-server service.

The first ten addresses and the last 54 addresses in the scope are reserved for purposes other than client allocation. The DHCP server also provides clients with default gateway and DNS server information.

On r4, which is connected to the client subnet through its eth0 interface, a DHCP helper is deployed to forward broadcast DHCP requests to the DHCP server. r4 runs the isc-dhcp-relay service and is configured to listen on eth0 and forward requests to 1.114.1.4.

This setup allows clients to obtain DHCP addresses even when the server is located in a different subnet.